

PHISHING DETECTION & AWARENESS REPORT

Cyber Security Internship Task 2

Future Interns

Repository: FUTURE_CS_02

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Date: August 2026

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1. Executive Summary

Phishing is a common form of social engineering in which attackers attempt to deceive users into revealing sensitive information, clicking malicious links, downloading harmful files, or making unauthorized payments.

This report analyzes three fictional phishing email samples representing common phishing scenarios: an account-lock warning, a fake delivery notification, and an unusual-login security alert. Each sample was examined for phishing indicators including urgency, fear-based language, suspicious sender information, requests for sensitive information, suspicious URLs, and pressure to take immediate action. All three samples were classified as high-risk phishing emails based on the combination of indicators identified.

The report also provides practical prevention guidelines and employee awareness recommendations to help users recognize and safely respond to phishing attempts.

2. Objective

The objective of this task is to analyze phishing email examples and identify characteristics that may indicate a phishing attempt.

The analysis focuses on:

- Identifying phishing indicators in email content.
- Examining sender information and suspicious domains.
- Identifying suspicious or misleading URLs.
- Recognizing social-engineering techniques such as urgency and fear.
- Classifying email risk.
- Explaining phishing risks in simple language.
- Providing practical recommendations to help users avoid phishing attacks.
- Creating an awareness resource that can help educate employees about phishing threats.

3. Tools Used

The following tools and resources were used during the project:

- Text editor Used to create and document the fictional email samples.
- Web browser Used for cybersecurity research and reviewing publicly available information.
- Microsoft Word Used to create the final professional report.
- GitHub Used to store and document the completed project and supporting evidence.

4. Analysis Methodology

Each email sample was analyzed using a structured process:

4.1. Sender Analysis

The sender address and domain were examined to determine whether the sender could be identified and verified.

4.2. Subject Analysis

The subject line was reviewed for urgency, threats, fear-based language, or unusual requests.

4.3. Content Analysis

The body of each email was examined for social-engineering techniques, requests for sensitive information, unusual instructions, and pressure to act quickly.

4.4. URL Analysis

Links included in the emails were examined as text for suspicious domains and unusual URL characteristics. Suspicious links were not opened.

4.5. Phishing Indicator Identification

The identified indicators were documented and considered together rather than relying on a single characteristic.

4.6. Risk Classification

Each email was classified based on the evidence identified during the analysis.

4.7. Recommended Action

Safe actions were provided to explain how a recipient should respond to each suspicious email.

Note: The email samples analyzed in this report are fictional examples created for cybersecurity education and demonstration purposes. They are designed to represent common phishing scenarios and were not sent to real recipients.

5. Phishing Email Analysis

5.1 Sample 1 — Account Lock Phishing

Subject: Urgent: Your Account Will Be Locked

Dear User,

We noticed suspicious activity on your account.

To avoid account suspension, please verify your details immediately.

Verify Now: [http://secure-account-verify\[.\]com](http://secure-account-verify[.]com)

Failure to verify within 24 hours will result in permanent account lock.

Regards,

Security Team

Analysis

1. Sender Analysis

Sender information: Not provided in the sample.

Assessment: The sender cannot be verified because no email address or domain is provided. This is a limitation of the sample, so sender legitimacy cannot be confirmed.

2. Subject Analysis

The subject creates urgency and fear by warning that the user's account will be locked.

Phishing emails commonly use urgent or threatening language to pressure recipients into acting without carefully checking the message.

Assessment: Suspicious.

3. Content Analysis

The email claims that suspicious activity has been detected and asks the recipient to verify their details immediately. The message uses fear and urgency to encourage the recipient to act quickly. It also threatens permanent account closure if the recipient does not respond within 24 hours. The greeting, "Dear User," is generic and does not identify the recipient by name.

Assessment: Highly suspicious.

4. URL Analysis

The email contains the following [URL:http://secure-account-verify\[.\]com](http://secure-account-verify[.]com)

The domain does not identify a known legitimate organization or service. The use of a generic security-related domain is suspicious, especially because the email asks the recipient to use the link to verify account information.

The URL uses HTTP rather than HTTPS. However, the lack of HTTPS alone does not prove that a website is malicious, so it should be considered one indicator rather than definitive evidence.

Assessment: Suspicious.

5. Phishing Indicators

The following phishing indicators were identified:

- Urgent and threatening subject line
- Fear-based language
- Request to verify account information
- Generic greeting
- Suspicious-looking domain
- Pressure to act within 24 hours
- Threat of permanent account closure
- Unexpected account-security request
- Link provided to perform the requested action

6. Risk Classification

Classification: PHISHING

Risk Level: HIGH

Reason: The email contains multiple social-engineering indicators, including urgency, fear, a generic greeting, a suspicious URL, and a request to verify account information. These characteristics strongly suggest an attempt to manipulate the recipient into clicking the link and potentially providing sensitive information.

6. Recommended Action

The recipient should not click the link or provide any personal information.

Instead, the recipient should:

- Verify the account status through the organization's official website or application.
- Avoid using links provided in unexpected emails.
- Report the email to the organization's IT or security team.
- Delete the message after reporting it.
- If the recipient already entered credentials, they should immediately change their password through the legitimate service and report the incident to the appropriate security team.

5.2 Sample 2 — Fake Delivery Notification

Subject: Delivery Attempt Failed – Action Required

From: Delivery Support support@parcel-delivery-alert.com

Dear Customer,

We attempted to deliver your package today, but your address could not be confirmed.

Please confirm your delivery information within 24 hours to avoid returning the package to the sender.

Confirm Delivery:

[http://parcel-confirmation\[.\]com/update](http://parcel-confirmation[.]com/update)

A small redelivery fee may be required.

Thank you,

Delivery Support Team

Analysis

1. Sender Analysis

Sender: Delivery Support <support@parcel-delivery-alert.com>

The sender uses a generic domain, "parcel-delivery-alert.com," rather than a clearly identifiable domain belonging to a known delivery company. The sender's identity cannot be independently verified from the email.

Assessment: Suspicious.

2. Subject Analysis

The subject, "Delivery Attempt Failed – Action Required," creates urgency and encourages the recipient to take immediate action. Delivery-related messages can be convincing because people may genuinely be expecting packages.

Assessment: Suspicious.

3. Content Analysis

The email claims that a delivery failed because the recipient's address could not be confirmed.

It asks the recipient to update delivery information within 24 hours and introduces the possibility of a redelivery fee. This combines urgency, a plausible situation, and a financial request to encourage the recipient to click the provided link.

Assessment: Highly suspicious.

4. URL Analysis

The email provides the following URL: [http://parcel-confirmation\[.\]com/update](http://parcel-confirmation[.]com/update)

The domain is generic and does not clearly identify a legitimate delivery company.

The URL also uses HTTP rather than HTTPS. However, the lack of HTTPS by itself does not prove that a website is malicious.

Assessment: Suspicious.

5. Phishing Indicators

The following indicators were identified:

- Urgent request for action
- 24-hour deadline
- Generic delivery-related sender domain
- Request to confirm personal information
- Suspicious URL

- Potential financial charge
- Generic greeting
- Attempt to create concern about a package
- Pressure to act through an email link

6. Risk Classification

Classification: PHISHING

Risk Level: HIGH

Reason: The email combines several common phishing techniques. It creates urgency, uses a potentially fake delivery identity, asks the recipient to provide information, directs the recipient to a suspicious website, and mentions a possible payment.

7. Recommended Action

The recipient should not click the link or provide personal or payment information. If the recipient is expecting a package, they should:

- Visit the delivery company's official website directly.
- Use the official tracking number if available.
- Contact the delivery company through its official customer-service channel.
- Report the suspicious email to the organization's IT or security team.
- Delete the email after reporting it.

5.3 Sample 3 — Fake Login Alert

Subject: Security Alert: Unusual Login Detected
From: Account Security security@account-protection-center.com
Hello,
We detected a login attempt from an unfamiliar device.
If this was not you, please secure your account immediately by confirming your identity:
[http://account-security-check\[.\]com/verify](http://account-security-check[.]com/verify)
You must complete verification within 12 hours or your account may be temporarily restricted.

Thank you,
Account Security Team

Analysis

1. Sender Analysis

Sender: Account Security security@account-protection-center.com

The sender uses a generic security-related domain rather than a clearly identifiable domain belonging to a known service provider. The sender's identity cannot be independently verified from the information provided.

Assessment: Suspicious.

2. Subject Analysis

The subject, "Security Alert: Unusual Login Detected," is designed to make the recipient concerned that their account may have been accessed by someone else. Security alerts can be legitimate, but unexpected alerts should be independently verified rather than acted on through links in the email.

Assessment: Suspicious.

3. Content Analysis

The email claims that an unfamiliar login was detected and asks the recipient to confirm their identity. It creates urgency by stating that verification must be completed within 12 hours and threatens temporary account restriction. This combination of fear, urgency, and a request to verify identity is a common social-engineering technique.

Assessment: Highly suspicious.

4. URL Analysis

The email contains the following URL: [http://account-security-check\[.\]com/verify](http://account-security-check[.]com/verify)

The domain is generic and does not clearly identify a legitimate service provider.

The URL also uses HTTP rather than HTTPS. However, the absence of HTTPS alone does not prove that a website is malicious.

The recipient should avoid using the link and instead access the relevant service through its official website or application.

Assessment: Suspicious.

5. Phishing Indicators

The following indicators were identified:

- Unexpected security alert
- Fear-based language
- Urgent 12-hour deadline
- Generic sender domain
- Request to confirm identity
- Suspicious URL
- Threat of account restriction
- Pressure to act through an email link

6. Risk Classification

Classification: PHISHING

Risk Level: HIGH

Reason: The email uses several social-engineering techniques to pressure the recipient into clicking a link and confirming their identity. The suspicious sender domain, urgent deadline, account-restriction threat, and verification request make the message highly suspicious.

7. Recommended Action

The recipient should not click the link or enter credentials or personal information.

Instead, they should:

- Open the relevant service through its official website or application.
- Check account activity directly.
- Change their password through the legitimate service if an unauthorized login is confirmed.
- Enable multi-factor authentication where available.
- Report the suspicious email to the organization's IT or security team.
- Delete the email after reporting it.

6. Risk Classification Summary

The three analyzed samples were classified as high-risk phishing emails based on the combination of indicators identified during the investigation.

Sample	Scenario	Classification	Risk Level
Sample 1	Account Lock Warning	Phishing	High
Sample 2	Fake Delivery Notification	Phishing	High
Sample 3	Unusual Login Alert	Phishing	High

The classification was based on multiple indicators rather than a single characteristic. The main indicators included urgency, fear-based language, suspicious sender domains, suspicious URLs, requests for information, and pressure to take immediate action.

7. Phishing Prevention Guidelines

Users can reduce the risk of falling victim to phishing by following these practices:

- Check the sender's email address carefully before responding.
- Be cautious of unexpected messages that create urgency or fear.

- Do not click links in suspicious or unexpected emails.
- Hover over links to inspect their destination before clicking.
- Verify important requests through the organization's official website or application.
- Never provide passwords, one-time passwords , or other sensitive information through an unexpected email.
- Be cautious of unexpected payment requests or requests for financial information.
- Do not open unexpected attachments.
- Report suspicious emails to the organization's IT or security team.
- If an account-security message appears suspicious, access the account directly through the official website rather than using the email link.
- Keep operating systems, browsers, and security software updated.

8. Employee Do's and Don'ts

Do:

- Verify unexpected requests independently.
- Check sender addresses and domains carefully.
- Inspect links before opening them.
- Use official websites or applications to access accounts.
- Report suspicious emails to the appropriate security team.
- Ask the IT or security team for assistance when uncertain.

Don't:

- Do not click suspicious links.
- Do not provide passwords or OTPs through email.
- Do not download unexpected attachments.
- Do not allow urgency or threats to pressure you into acting quickly.
- Do not assume an email is legitimate simply because it uses company logos or professional-looking formatting.

- Do not reply to suspicious messages asking for sensitive information.

9. Conclusion

The analysis of the three email samples demonstrates how phishing attacks use social-engineering techniques to manipulate users into taking unsafe actions.

The samples used several common phishing indicators, including urgency, fear, suspicious sender information, suspicious URLs, requests for verification, and threats of account restriction or financial consequences.

Recognizing these indicators can help employees stop phishing attacks before they result in credential theft, financial loss, or unauthorized access.

Users should avoid acting directly on unexpected requests and should instead verify information through trusted and official channels. Security awareness and timely reporting are important parts of an organization's overall defense against phishing.

10. References

- Future Interns — Cyber Security Task 2: Phishing Email Detection & Awareness System.
- Google Safety Center — Phishing and suspicious message awareness resources.
- Public phishing email repositories referenced in the Future Interns task instructions for educational research.